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CN Lab Project (Section A)

Financial Institution System Networking Project

Introduction

Jubilee Financial Services Ltd (JFSL) is a prominent finance service provider in Kenya, headquartered in an eleven-story building in Nairobi. This report details the design and implementation of the network infrastructure for JFSL, aiming to meet the organization's needs for security, performance, scalability, and availability.

The proposed solution ensures a robust, secure, and redundant Local Area Network (LAN) and Wide Area Network (WAN) infrastructure, leveraging modern technologies like VLANs, OSPF, DHCP, and VoIP to enhance functionality and security.

Project Scope

Departments and Device Allocation:

Seventh Floor: HR, Customer Service (CS), and Marketing (MK) with 40 devices, 40 IP phones, and a WiFi-AP per department.

Eighth Floor: Legal Management (LM) and IT with 20 devices, 20 IP phones, and a WiFi-AP per department.

Network Infrastructure Requirements:

VLANs to segregate departmental traffic for enhanced security and performance.

Separate voice VLANs (VID 120) for IP phones.

Hierarchical design model with redundancy at each layer.

WAN Connectivity:

Connections to Safaricom and JTL ISPs for internet redundancy.

Secure communication to external servers hosting DHCP, DNS, WEB, and EMAIL services.

Security Policies:

Use of Access Control Lists (ACL) to control user access.

VPN configuration for secure remote communication.

Network Design

Topological Overview

The network design is based on a hierarchical model, providing:

Core Layer: Comprising multilayer switches for routing and switching.

Distribution Layer: Ensuring VLAN and inter-VLAN routing with high availability.

Access Layer: Comprising access switches for user connectivity.

IP Addressing				
Base Network: 192.168.20.0				
HQ Network				
Department	Network & Subnet Mask	Valid Host Addresses	Default Gateway	Broadcast Address
HR	192.168.20.0/26	192.168.20.1 to 192.168.20.62	192.168.20.1	192.168.20.63
CS	192.168.20.64/26	192.168.20.65 to 192.168.20.126	192.168.20.65	192.168.20.127
MK	192.168.20.128/26	192.168.20.129 to 192.168.20.190	192.168.20.129	192.168.20.191
LM	192.168.20.192/27	192.168.20.193 to 192.168.20.222	192.168.20.193	192.168.20.223
IT	192.168.20.224/27	192.168.20.225 to 192.168.20.254	192.168.20.225	192.168.20.255

```
'HQMLSW1'(config-if)#exit
'HQMLSW1'(config)#ip routing
'HQMLSW1'(config)#
'HQMLSW1'(config)#router ospf 10
'HQMLSW1'(config-router)#router-id 1.1.1.1
'HQMLSW1'(config-router)#network 10.10.10.0 0.0.0.3 area 0
'HQMLSW1'(config-router)#
'HQMLSW1'(config-router)#network 192.168.21.16 0.0.0.3 area 0
'HQMLSW1'(config-router)#network 192.168.20.0 0.0.0.255 area 0
'HQMLSW1'(config-router)#exit
'HQMLSW1'(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
'HQMLSW1'(config)#
```

Technologies Implemented

Device Configuration:

Hostnames, passwords, banner messages, and IP domain lookup disabling.

VLANs and Inter-VLAN Routing:

Configured VLANs for each department.

Configured inter-VLAN routing using multilayer switches.

Dynamic IP Addressing:

Dedicated DHCP server for data VLAN.

DHCP configuration on routers for voice VLAN.

Routing Protocol:

OSPF configured for dynamic route advertisement.

VoIP Configuration:

Cisco 2811 router configured for telephony services and VoIP.

```
Enter configuration commands, one per line. End with CNTL/Z.
'HQMLSW1'(config)#interface GigabitEthernet1/0/1
'HQMLSW1'(config-if)#no switchport
'HQMLSW1'(config-if)#ip add 192.168.21.17 255.255.255.252
'HQMLSW1'(config-if)#no shut
'HQMLSW1'(config-if)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
'HQMLSW1'(config-if)#
```

```
Enter configuration commands, one per line. End with CNTL/Z.
'HQMLSW2'(config)#interface GigabitEthernet1/0/1
'HQMLSW2'(config-if)#exit
'HQMLSW2'(config)#
'HQMLSW2'(config)#ip routing
'HQMLSW2'(config)#router ospf 10
'HQMLSW2'(config-router)#router-id 2.2.2.2
      ^
% Invalid input detected at '^' marker.

'HQMLSW2'(config-router)# router-id 2.2.2.2
'HQMLSW2'(config-router)#network 10.10.10.0 0.0.0.255 area 0
'HQMLSW2'(config-router)#
'HQMLSW2'(config-router)#network 192.168.21.20 0.0.0.3 area 0
'HQMLSW2'(config-router)#
'HQMLSW2'(config-router)#network 192.168.20.0 0.0.0.255 area 0
'HQMLSW2'(config-router)#exit
'HQMLSW2'(config)#do wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
'HQMLSW2'(config)#
```

Implementation Details

Subnetting and IP Addressing:

Data VLAN: 192.168.20.0/24

Voice VLAN: 10.10.10.0/24

Public Addresses: 190.200.100.0

Redundancy and Security:

Redundant connections to ISPs for load balancing and failover.

Port security configured on switches using sticky MAC addresses.

Testing and Validation:

Ensured communication between devices in all departments.

Verified internet connectivity and secure remote access using SSH.

Server-side Site					
No.	Branch	Network & Subnet Mask	Valid Host Addresses	Default Gateway	Broadcast Address
1	Server-Side LAN	192.168.21.0/28	192.168.21.1 to 192.168.21.254	192.168.21.1	192.168.21.15

Between the Routers and Layer-3 Switches	
No.	Network Address
HQR- HQMLS1	192.168.21.16/30
HQR- HQMLS2	192.168.21.20/30

Between the Routers and ISPs- 190.200.100.0	
No.	Network Address
HQR- ISP1	190.200.100.0/30
HQR- ISP2	190.200.100.4/30
SVR- ISP1	190.200.100.8/30
SVR- ISP2	190.200.100.12/30

```

Password:
Password:
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface Serial0/2/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/2/0
Router(config-if)#exit
Router(config)#router ospf 10
Router(config-router)#router-id
% Incomplete command.
Router(config-router)#router-id 5.5.5.5
Router(config-router)#network 190.200.100.0 0.0.0.3 area 0
Router(config-router)#
02:03:03: %OSPF-5-ADJCHG: Process 10, Nbr 4.4.4.4 on Serial0/2/0 from LOADING to FULL,
Loading Done

```

Results

Successful deployment of VLANs with distinct subnet allocations. Functional inter-VLAN routing, DHCP, and VoIP communication. Secure and redundant network infrastructure meeting all design requirements.

```
Router(config)#interface FastEthernet0/0
Router(config-if)#no ip add
Router(config-if)#
Router(config-if)#interface FastEthernet0/0.120
Router(config-subif)#
Router(config-subif)#encapsulation dot1Q 120
Router(config-subif)#ip add 10.10.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
```

Conclusion

The designed network infrastructure for JFSL meets all requirements for security, scalability, and high performance. The implementation ensures seamless connectivity across all departments and secure communication with external servers, aligning with the organization's objectives.

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